Assignment 2: Coordinate systems & GNSS

CE334 Modern Methods in Geoinformatics 2025-26 II
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1. List the major GNSS constellations currently in operation. Prepare a comparative table including the number of satellites, orbital characteristics, signal frequencies and frequency bands used by each constellation.
2. Describe the major error sources in GNSS positioning, with special emphasis on ionospheric and tropospheric delays. Mention the typical range of positioning errors caused by these effects and explain why they vary.
3. What are precise satellite orbits in GNSS? Explain their role in improving positioning accuracy and mention applications where precise orbits are essential.
4. Using Newton's Laws of Motion and Universal Gravitation, derive the equation of a satellite's orbit and prove that the trajectory is an ellipse.
5. Explain how CDMA (Code Division Multiple Access) is utilized in GNSS systems to enable multiple satellites to transmit on the same frequency band without interference. Discuss the role of PRN codes in this process and why is it critical for satellite identification? Explain how the PRN code is used to calculate the time of signal travel.

Bonus (+20%)

Explain the transformation of coordinates between the Earth-Centered Earth-Fixed (ECEF) and Earth-Centered Inertial (ECI) reference frames. Highlight the role of Earth's rotation in converting coordinates from ECEF to ECI and vice versa.

Note : Submit handwritten copy.